

Solution Requirements

Date	15 July 2026
Team ID	XXXXXX
Project Name	StudyAI – Smart AI educate Companion & Quiz Generator
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Users can securely register and log in using email and password. Passwords are encrypted, and email verification is required.</i>	High
2	Authorization levels	<i>The system provides role-based access. Students can upload materials and access AI features. Administrators can manage users, documents, and system settings.</i>	High

3	External interfaces	<i>The system integrates with Google Gemini API/Groq API for AI content generation and Supabase for authentication, database, and storage services.</i>	High
		<i>services the system must connect to]</i>	
4	Transactions processing	<i>The system processes document uploads, AI-generated summaries, quizzes, flashcards, and study schedules and stores them in the database.</i>	High
5	Reporting	<i>The system generates analytics dashboards showing study progress, quiz scores, weak topics, and learning statistics.</i>	medium
6	Business rules, etc.	<i>Only authenticated users can access study materials. Quiz and flashcard generation require uploaded documents. Generated content is linked to the user's account.</i>	High
7	Compliance to laws or regulations	<i>The system follows data privacy and security standards, protecting user information and complying with educational data protection guidelines.</i>	medium
8	Other	<i>The system provides AI-based personalized study recommendations and study plans based on user performance.</i>	medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>The system should quickly load pages and generate AI responses efficiently.</i>	<i>Page loads in less than 2 seconds and AI responses generated in less than 10 seconds.</i>
2	Scalability	<i>The system should support an increasing number of users and study materials without performance degradation.</i>	<i>Supports 10,000+ registered users and 500 concurrent users.</i>
3	Security & Data Privacy	<i>User data should be encrypted and protected from unauthorized access.</i>	<i>Password encryption, HTTPS communication, and secure authentication.</i>
4	Reliability & Availability	<i>The system should remain accessible with minimal downtime.</i>	<i>99.9% uptime per month.</i>
5	Usability & Accessibility	<i>The interface should be easy to use and accessible on different devices.</i>	<i>Responsive design and user-friendly interface with minimal learning curve.</i>
6	Other	<i>The system should allow easy updates and addition of new features.</i>	Modular codebase with proper documentation and reusable components.